



A generative design method for structural topology optimization via transformable triangular mesh (TTM) algorithm

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Abstract

This article presents a way of optimizing the conduction topology for heat-generating structures by means of transformable triangular mesh (TTM) algorithm which is implemented in an explicit and geometrical way. Unlike the traditional optimization approaches, the proposed method capitalizes on the use of a special morphing algorithm to generate optimal topologies from a genus zero surface. In this method, the initial geometry is firstly converted into triangular mesh and stored as a half-edge data structure. Then, the mesh operations (i.e., subdivision, split, and refinement) are employed to activate the geometry to move, split, and deform upon the underlying finite element mesh so that the conduction topology can be achieved by optimizing the positions and orientations of the triangular grids. The unique feature of the mesh operation is the split, which makes the geometries have different number of faces, edges, vertices as the initial one, and therefore different genus number between these geometries. This method renders the optimization process more flexibility. Finally, some examples with verification results are presented to demonstrate that TTM algorithm is capable of proposing solutions having almost the same cooling effectiveness with less computing resources compared with the commonly used density approaches.

Keywords Topology optimization · Heat conduction · Transformable triangular mesh (TTM) · Mesh processing

1 Introduction

Topology optimization design is an established optimization design technology in the field of structural design. It is a powerful tool in the conceptual design of a product, and is used to find the proper material distribution under the prescribed conditions to achieve optimal structural performance. Since the proposition of the homogenization design method (HDM) (Bendsøe and Kikuchi 1988) in 1988, topology optimization research has received extensive attention. Topology optimization methods can be divided into the following types: (a) density-based method, including the solid isotropic material with penalization (SIMP) method (Rozvany and Bendsee 1995) and the rational approximation of material properties (RAMP) method (Yan et al. 2018); (b) level set method (Wei

et al. 2018); (c) topological derivative method (de Faria and Lesnic 2015); (d) evolutionary method, including evolutionary structural optimization (ESO) method and the bidirectional ESO method (Huang and Xie 2008); (e) phase field method (August et al. 2015); and (f) deep learning-based topology optimization method (Xin et al. 2019; Li et al. 2019a). Among the density-based methods, the representative method is the SIMP method. It is worth noting that the SIMP method has been widely used in commercial software. In this method, the design domain is first discretized into finite elements (i.e., pixels) with reasonable resolution. Then, the elementwise black and white density distribution is determined through numerical analysis and optimization, which represents the topology of the structure. The density of pixels is between 0 and 1; pixels with density of 0 are treated as blank areas; pixels with density of 1 are treated as solid areas, and gray parts with densities between 0 and 1 are eliminated as much as possible by the penalty method (Martínez 2005). Finally, the optimized structure is represented by the distribution of black and white pixels. Although the method has achieved remarkable results, there are still some problems to explore. First, the method uses pixels to form an implicit representation of the boundary, and thus, this description framework is different from that of the

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computer-aided design (CAD) platform, which describes a boundary using explicit expressions of geometric parameters (such as vertices, edges, faces, and their Boolean operations). Second, because this pixel-based description lacks geometric information, the feature size of the optimized structure is difficult to precisely control, which brings difficulties to manufacturing (Sigmund 2009). Finally, because the method uses pixels to represent the topology, a higher pixel resolution is required when clear boundaries are needed, which results in a relatively high computational cost, especially in large-scale design (Aage et al. 2017).

Let us consider the level set method as an example to compare with the SIMP method. The SIMP method optimizes the density of materials at the microscopic level to form the macroscopic topological structure, whereas the focus of the level set method is boundary evolution. This method also discretizes the design domain into finite elements for numerical analysis to calculate structural performance. The isocontours of the level set function are typically used to implicitly express the boundaries of the optimized structure. Compared with the SIMP method, the boundary representation of the level set method is more concise. Although geometric information such as boundary curvature can be calculated from the level set function, its boundaries are only geometrically described rather than parametrically, and therefore, the method is also implicit. The conventional level set method needs to solve the Hamilton–Jacobi equation and needs to re-initialize the level set function to ensure the stability of the optimization, which makes the solution inefficient. Wang et al. (Wang and Wang 2006) used the radial basis function (RBF) as the interpolation function to combine RBF with level set method. This method converts the Hamilton–Jacobi partial differential equation into an ordinary differential equation, thereby reducing the complexity of the solution. The RBF can meet the continuity and smoothness requirements during the topology optimization process, so there is no need to re-initialize the level set function. Luo et al. (2008) introduced compactly supported radial basis function (CS-RBF) into the level set method and proposed the parametric level set method (PLSM), which effectively solved the problems existing in the conventional level set method.

In recent years, researchers have proposed a series of explicit geometric description frameworks (Mei et al. 2008; Zhou et al. 2016; Norato et al. 2015). In 2014, Guo et al. (2014) proposed a representative explicit method called the moving morphable component (MMC) method, in which a set of moving morphable components are used as building blocks for topology optimization. The optimal structure is formed by optimizing the shape and layout of these components and utilizing the overlapping and merging of these components. Further theoretical study and implementation developed of this method can be found in Li et al. (2019b, 2020). The geometrical description framework of the MMC-like

methods can be considered as a double-layer architecture. The upper layer comprises parametrically controlled moving morphable components that represent the geometry of the optimized structure. The bottom layer is a finite element mesh (FEM) by which physical properties (such as elastic modulus, thermal conductivity, and electrical conductivity) are calculated. The boundary of the components is projected from the upper layer to the bottom layer in each iterative step. The properties of the FEM nodes inside the projected boundary are set as solid materials, and those outside the projected boundary are set as weak materials. This property information is inserted into the finite elements, and the physical properties of the finite element mesh are calculated by interpolation. The components are driven to achieve optimal movement and deformation results by introducing the sensitivity of the constraints and objectives to the moving asymptote (MMA) (Svanberg 1987) optimizer for updating the variables in the optimization iteration. After the termination condition is triggered, the above iterative mode stops. In this way, MMC-like methods can construct an optimal topology by using multiple explicit components described by parameters.

In geometric topology, the number of holes in a geometry is called its genus (Zhang et al. 2012). Because shape optimization methods usually transform between geometries with the same genus, changes in the genus become the essential difference between shape optimization and topology optimization methods (Bendsoe and Sigmund 2013). In MMC-like methods, the geometry of the optimized structure is formed by the movement, deformation, and overlap of multiple components, where the overlap of multiple components is the core operation for changing the genus of the structure. However, this operation requires a certain number of components to form holes in the optimized structure, and when the number of components is insufficient, it is difficult to form a global optimum structure. This uncertainty makes the optimization results highly dependent on the initial components, and a large number of initial components is required to achieve good optimization results. To reduce the impact of the initial component layout on the optimal structure, it is necessary to consider how to use only a simple initial component layout to form the global optimal structure efficiently and robustly.

The morphing technique in computer graphics community aims to create a smooth transition from one initial graphical object to a target graphical object with different geometric shapes (Beier and Neely 1992; Jacobson et al. 2011; Zhou et al. 2014), where triangular meshes are used to describe the graphical object. The “smooth” means that triangular mesh faults such as non-manifold edges and self-intersections (Li and Iyengar 2015; Baek et al. 2015; Wang et al. 2017) do not occur during morphing. The morphing process first selects vertices (also known as hot vertices) from the initial geometry boundary and then moves these hot vertices to the corresponding vertices on the boundary of the target geometry to create

smooth morphing of the initial geometry into the target geometry. These hot vertices, under the control of the constraints, can drive the entire triangular mesh to deform accordingly (Alexa et al. 2000; Igarashi et al. 2005) and maintain the original appearance of the in-between mesh as much as possible, which results in a natural visual effect. The smooth transition in this technique between geometries of different shapes inspired us.

It is worth noting that there are a number of mesh processing operations in the computer graphics community that can be utilized, such as mesh subdivision, mesh split, and mesh refinement. The mesh subdivision (Rossignac 1999) operation can divide an original polygon into several smaller polygons, which increases the number of polygons in the mesh under certain quality criteria. This operation also increases the number of geometric primitives in the mesh (vertices, edges, faces), which increases the degree of freedom of the mesh, making the morphing more natural. The mesh split (Kovacs et al. 2015) operation can change the connection relationship between polygons in the mesh. When the connection between two adjacent internal polygons is broken, one potential hole can be generated in the mesh, which increases the genus of the mesh. If this happens to polygons on the boundary, it will cause a Y-branch, which adds extra freedom of deformation. The mesh refinement (Patterson et al. 2015; Areias et al. 2018) operation can reconstruct a geometry. The polygons in the reconstructed geometry should be as similar as possible, which is consistent with the basic criterion for smooth morphing. In smooth morphing, the sum of the movements of all hot vertices should be as small as possible, which requires that the amount of change in mesh size caused by the movement of each hot vertex should be similar. However, the amount of change in mesh size caused by the movement of the hot vertex in a larger polygon is larger than that caused by the movement of the hot vertex of a smaller polygon. Therefore, the hot vertices of larger polygons are more likely to change the properties of the mesh than the hot vertices of small polygons, which makes the morphing unnatural. When the polygon in the mesh changes too much and the mesh becomes distorted, the mesh refinement operation can restore the mesh to its regular shape (Chan et al. 2016).

The natural morphing of geometry in geometric morphing technology has inspired us. By combining geometric morphing technology and the topology optimization method, a method called transformable triangular mesh (TTM) is proposed in this work, which is also a two-layer architecture. The upper layer uses the triangular mesh described by the half-edge data structure (Campagna et al. 1998) as the component to represent the geometry of the optimized structure, and the bottom layer is the FEM used for calculation and analysis. With this method, the shape and genus of the triangular mesh can be changed through adaptive mesh processing operations, and thus efficient and robust convergence of global optimality can be achieved by using only one zero-genus component,

which reduces the dependence of the optimization results on the initial component layout.

In 2017, Lian and Christiansen et al. proposed a DSC optimization method (Lian et al. 2017) in which the design domain is discretized into triangular elements. These elements are used both for structural representation and for finite element analysis. By introducing an accurate p-norm, the resulting structure of the method can have an explicit smoothing boundary. This method provides a very useful idea.

Also starting from the mesh operation to innovate the topology optimization method, the TTM method provides a different idea. The TTM method separates the structural description from the finite element calculation, using a zero-genus explicit triangular mesh as the component to describe the optimized structure and provides a set of flexible and adaptive mesh operations, while the fixed finite element mesh is only used for finite element analysis. The TTM method has four advantages. First, the TTM method studies the mesh operation of the Lagrangian mesh, which is easier and more flexible than the mesh operation of Euler mesh. Second, in the TTM method, the number of variables is independent of the finite element mesh, so a fine finite element mesh can be used to increase the calculation accuracy. Third, the number of initial variables in the TTM method is small. As the iteration proceeds, the adaptive mesh operations increase the number of variables according to the optimization trend, which makes the calculation cost increase reasonable. Fourth, the highly flexible component reduces the dependence of the optimization result on the initial component layout. The TTM method provides a completely different way of mesh operation and topology addition from the DSC optimization method, which can provide a different idea for subsequent researchers.

The rest of this paper is organized as follows. Section 2 introduces the idea and mesh operation of the TTM framework in topology optimization and the topology optimization problem formula based on this framework. Section 3 presents the numerical analysis and interactions between the TTM and FEM layers. Section 4 discusses five case studies using the TTM algorithm to realize the optimal structural layout and presents the comparison of the simulation and experimental results. Finally, Sect. 5 summarizes the paper.

2 Transformable triangular mesh-based topology optimization framework

In this section, the basic idea of the proposed TTM-based topology optimization approach is explained in detail.

2.1 Geometric morphing

In computer graphics, the morphing operation refers to the smooth transition between two geometric models (i.e., source

model and destination model). There are two basic issues: (1) How points on one object get mapped to points on another object (i.e., correspondence) and (2) how an object transitions to a new object (i.e., interpolation).

Take the process of continuous change from geometry “I” to geometry “C” as an example to illustrate the characteristics of smooth geometric morphing, as shown in Fig. 1a. First, several vertices are selected as hot vertices on the boundary of the initial geometry I and corresponded to the pre-set hot vertices on the boundary of the target geometry C. Then, we move each hot vertices of the initial geometry I to the corresponding hot vertices of the target geometry C simultaneously, and thus, the geometry is completely transformed from I to C. During this deformation process, the genus of the geometry remains constant. However, in some cases, the genus changes with deformation. When the initial and target geometries have different genera, we say that there is a difference in the number of holes on the geometry. Morphing between such geometries with different genera is often unnatural or even accompanied by self-intersection.

We consider the morphing of geometry I to “B” depicted in Fig. 1b, as an example of the morphing of different genera. The initial geometry I has no hole, whereas the target geometry B has two holes. Therefore, the genera of I and B are, respectively, 0 and 2. Without hole boundaries, there are no internal hot vertices in the initial geometry I to correspond to the internal vertices of the target geometry B. If the morphing is forced, the geometry will distort and self-intersect during the morphing process due to the transition from an external boundary to a hole boundary.

The common way to solve this problem is to manually introduce pseudo internal boundaries and set hot vertices inside the initial geometry I to correspond to the internal vertices of the target geometry B. Thus, the deformation of variable

genera is realized smoothly and naturally without self-intersection, as shown in Fig. 1c. In this way, if we know how and where to place the hot vertices, natural deformation between the two geometries can be obtained regardless of their topological information.

The method of adding pseudo internal boundary vertices inside the geometry enables a smooth transition between geometries with different genera, which provides a good idea for topology optimization based on the transformable triangular mesh proposed in this work.

2.2 Mesh description

In recent years, mainstream explicit topology optimization methods often control multiple simple components. The design variables in the optimization method are often parameters of each component’s size and position. Through the overlap and concealment of components, an optimized structure is obtained after multiple iterations. However, in the present work, only one component is controlled, i.e., the TTM, instead of multiple ones. The design variables in this method are the coordinates of the control vertices (i.e., hot vertices) in the mesh, which are combined with subsequent mesh operations to enable the mesh to move, deform, and change its own genus rather than simply changing the sizes and positions of the components. Therefore, the explicit topology optimization method proposed in this study can form an optimized structure with only one component. To realize topology optimization, we need to parameterize the geometric information.

In the first step of parameterization, we triangulate the geometry, so that the transformable triangular mesh is described by a number of adjacent triangular grids whose vertices and edges are shared by their adjacent triangular grids. Then, a mesh description is required to save the geometry data for

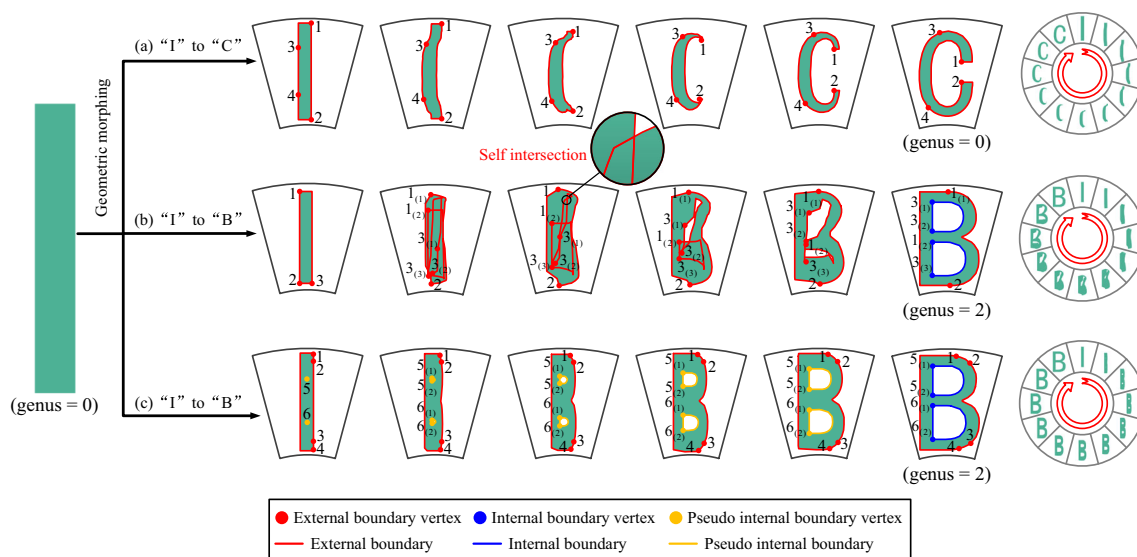


Fig. 1 Geometric morphing process without and with pseudo internal boundary vertices

subsequent mesh operations. This paper introduces a half-edge data structure to store mesh data, including vertices, edges, and faces of all triangular grids, which allows us to change vertex coordinates and maintain or break the connection between the triangular grids.

The half-edge data structure uses half edges to store the geometric data and connection information of the mesh, including the starting and ending vertices of each half edge and its next, previous, and opposite half edges. In addition, the face to which the half edge belongs is also stored. This data structure allows any half edge of the entire mesh to be traversed in the mesh without topological errors. The triangle grids and their half-edge data structure are shown in Fig. 2a. These geometry and connection information form the basis for the design variables of the topology optimization.

During the iterations, all the boundary edges need to be determined so that the vertices on the boundary can be selected as the deformation control vertices. Because of the half-edge data structure, it is easy to traverse all the half edges of the mesh to extract the mesh boundaries. The traversal process is illustrated in Fig. 2b. As can be seen from the figure, the half edge of the mesh boundary has no opposite half edge. During the traversal process, a half edge that meets the above criteria needs to be identified first, set as the boundary edge, and marked as HE_1 . Next, start with this

half edge and traverse in the order of $HE_1 \rightarrow next \rightarrow opposite$ until the next boundary HE_2 is determined and the traversal is interrupted. Then, HE_2 is used as the new starting half edge, and the above steps are repeated. This traversal process stops after all the half edges of the mesh boundary are marked end to end.

Finding the vertices on the boundary is preparation work for mesh deformation, and the deformation consists of four main elements: (1) the triangulated mesh described in the half-edge data structure, (2) one or more vertices on the boundary that can actively change (i.e., control vertices), (3) regions in which control vertex can drive other non-control vertices to change together (i.e., the region of interest, ROI), and (4) the target position of each control vertex. To better control the changes of non-control vertices in the ROI, we use the “as rigid as possible shape deformation” method. The basic deformation form is shown in Fig. 2c.

The Laplacian coordinate (Botsch et al. 2010) of the i th vertex in triangular grid $\mathbf{M}$, proposed to describe the movement of the shape, measures a vertex's weighted distance to all connected vertices. The Laplacian coordinate can be calculated as

$$\mathbf{L}(\mathbf{v}_i) = \sum_{\mathbf{v}_j \in \mathbf{N}(\mathbf{v}_i)} w_{ij} (\mathbf{v}_i - \mathbf{v}_j) \quad (1)$$

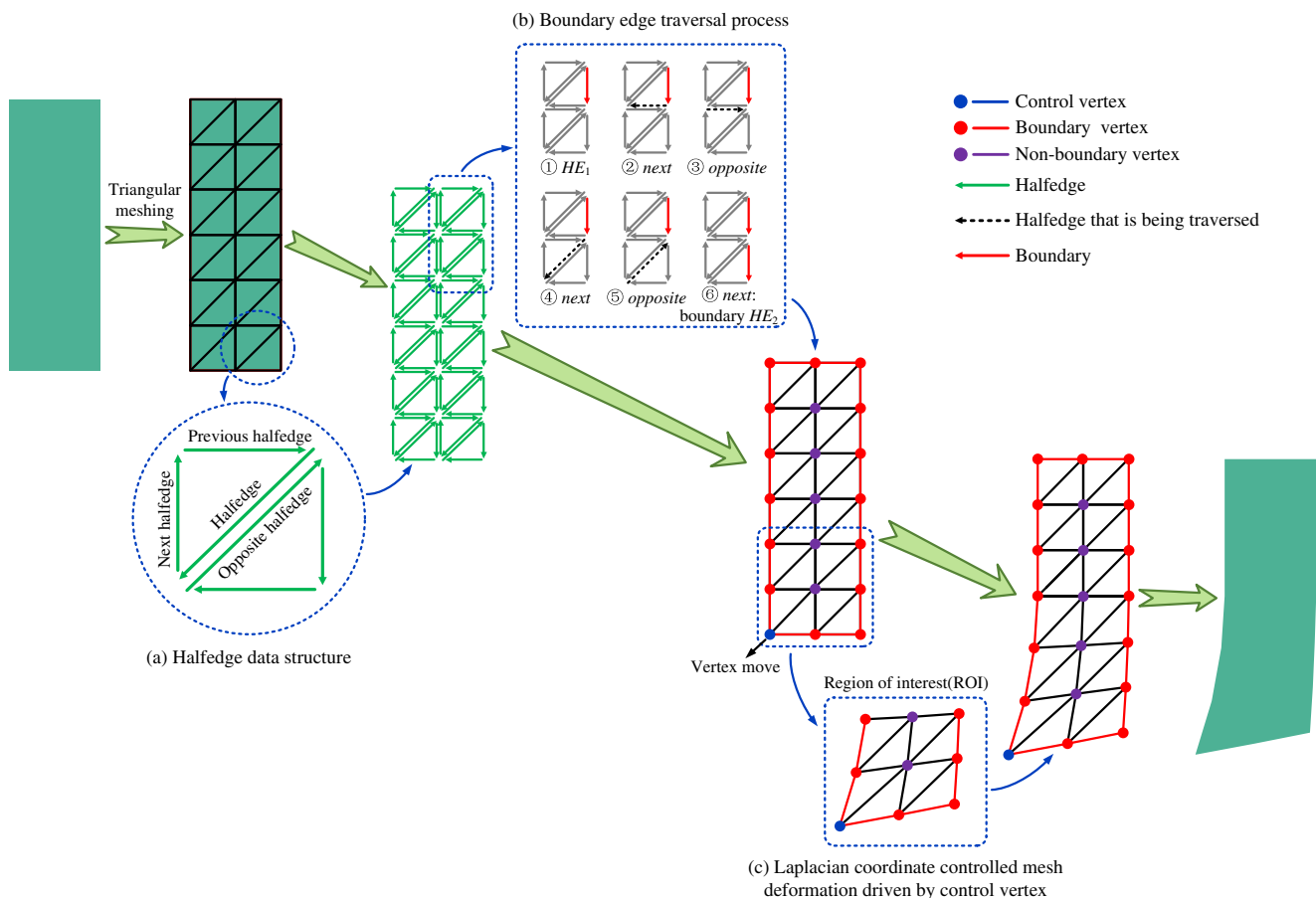


Fig. 2 Geometric morphing using triangular grid and half-edge data structure

where $\mathbf{N}(\mathbf{v}_i)$ represents the vertices connected to the i th vertex, $\mathbf{v}_i$ denotes the Cartesian coordinate of the i th vertex, $\mathbf{v}_j$ denotes the Cartesian coordinate of the vertex that is connected to the j th vertex, and w_{ij} denotes the weight for the edge $\mathbf{v}_i\mathbf{v}_j$. The sum of w_{ij} equals 1, and the weight used is the cotangent scheme (Levi and Gotsman 2014). The cotangent weight corresponding to the edge of the surface mesh is the average of the cotangents of the angles opposite the edge, and thus, the results obtained are not subject to the surface mesh deviation of the approximate surface (Igarashi et al. 2005).

The entire Laplacian coordinates should be minimized for optimal deformation as a control vertex's movement is given. The energy function E can be calculated from Eq. (2).

$$E = \sum_{\mathbf{v}_i \in \mathbf{M}} \sum_{\mathbf{v}_j \in \mathbf{N}(\mathbf{v}_i)} w_{ij} \left\| \left(\mathbf{v}'_i - \mathbf{v}'_j \right) - \mathbf{R}_i (\mathbf{v}_i - \mathbf{v}_j) \right\|^2 \quad (2)$$

In this equation, $\mathbf{R}_i$ is a 3×3 rotation matrix, w_{ij} is the weight for the edge $\mathbf{v}_i\mathbf{v}_j$, and $\mathbf{v}'_i$ denotes the Cartesian coordinates of the i th vertex after movement. The as rigid as possible shape deformation is defined by minimizing the energy function E .

In Eq. (2), the new positions $\mathbf{v}'_i$ and $\mathbf{v}'_j$ and the rotation matrix $\mathbf{R}_i$ are unknown. Because the energy contribution of each vertex is positive, it is only necessary to minimize the energy contribution of each vertex $\mathbf{v}_i$ and $\mathbf{v}_j$ to achieve the goal of minimizing the energy function E . This minimization can be implemented in two steps. First, the translation of each vertex is treated as zero to obtain an approximate rotation matrix. Second, the rotation matrix $\mathbf{R}_i$ is fixed and the new positions $\mathbf{v}'_i$ and $\mathbf{v}'_j$ are calculated. This turns the original problem into a linear least squares problem, which only requires the solution of the linear system of the partial derivatives of the above equations. Assuming that there are symmetric weights, the i th equation of the linear system can be written as

$$\sum_{\mathbf{v}_j \in \mathbf{N}(\mathbf{v}_i)} w_{ij} (\mathbf{v}'_i - \mathbf{v}'_j) = \sum_{\mathbf{v}_j \in \mathbf{N}(\mathbf{v}_i)} w_{ij} \frac{(\mathbf{R}_i + \mathbf{R}_j)}{2} (\mathbf{v}_i - \mathbf{v}_j) \quad (3)$$

When we use cotangent weights, the above equation can be modified to Eq. (4).

$$\begin{aligned} \sum_{\mathbf{v}_j \in \mathbf{N}(\mathbf{v}_i)} (w_{ij} + w_{ji}) (\mathbf{v}'_i - \mathbf{v}'_j) \\ = \sum_{\mathbf{v}_j \in \mathbf{N}(\mathbf{v}_i)} (w_{ij} \mathbf{R}_i + w_{ji} \mathbf{R}_j) (\mathbf{v}_i - \mathbf{v}_j) \end{aligned} \quad (4)$$

This two-step optimization is iteratively applied several times to obtain a better result.

2.3 Mesh transformation control

To realize topology optimization explicitly, the mesh should be able to perform not only geometric deformation but also

connectivity transformations. In this study, three different transformation operations are considered: mesh subdivision, split, and refinement.

The topology optimization framework is divided into two aspects: The fixed background mesh is used for finite element calculation and the component represents the geometry of the optimized structure. In this framework, a triangular mesh is used as the component. Because the mesh is composed of many triangular grids with high degrees of freedom, the mesh itself has strong deformation ability. Guided by the information fed back by the background mesh, the component further needs to have higher deformability to achieve the optimal structure. More triangular grids of the triangular mesh are required, and the triangular mesh is densified by the subdivision operation. If the component's deformation ability cannot meet deformation requirements, the component needs some new branches or holes that can be formed by the split operation. As topology optimization proceeds, the component may be subjected to distortions such as overlapping. The refinement operation can change the distortional mesh into a regular one, after which the topology optimization can work properly. The principles of these three operations are explained in detail below.

2.3.1 Subdivision

The geometry of the structure becomes more complex with topology optimization, which requires the mesh to densify itself and generates more control points to increase its deformability. In this work, the primal triangle quadrisection (PTQ) (Oswald and Schröder 2003) is used to implement mesh subdivision. As seen in Fig. 3a, the subdivision action is triggered when the length of the edges of any triangle grid exceeds a preset threshold L_s . Subdivision of mesh is implemented by dividing the original triangle into four blocks. First, the three midpoints of the original grid's sides are created as three new vertices. Next, a triangular grid with these new vertices, as well as the original ones, is reconstructed, and thus, a densified mesh is obtained.

The subdivision operation enhances the deformability of the mesh to a certain extent, which can make the mesh deformation more natural.

2.3.2 Split

The split operation has two purposes. First, it creates new boundaries to increase the deformability of the mesh. When a split occurs at the edge of the triangular mesh, it becomes a branching process by which the triangular mesh becomes more flexible. Second, it increases the genus of the mesh. Compared with excessive deformation and self-intersection of the mesh, the easiest way to increase the genus in a geometry is to create holes inside the mesh, which creates new

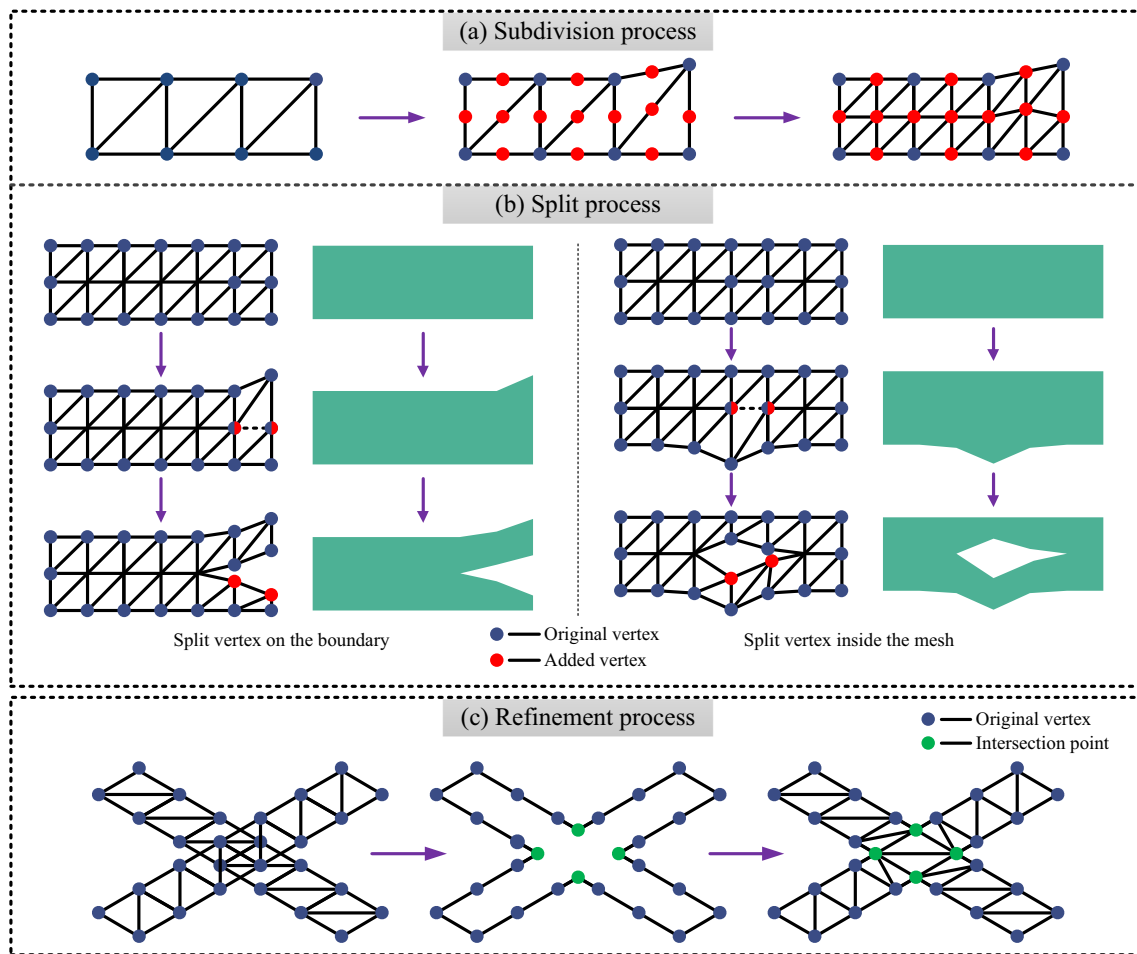


Fig. 3 Three different transformation operations in the TTM framework

internal boundaries. More specifically, the aforementioned half-edge data structure has connection properties between adjacent triangle grids, i.e., the two adjacent triangular grids have a shared edge, so we have to break this connection by turning the vertex shared between adjacent grids into a number of independent vertices and reconstructing the adjacent grids with these vertices. In brief, the process is implemented by copying the vertices shared by neighboring triangular grids and reconstructing the triangular grids. In the half-edge data structure, the number of each triangular grid is saved. For example, the m th grid and the n th grid are adjacent triangular grids (number $m < n$), and the distance between two non-shared vertices in the two grids is D_{mn} . When D_{mn} exceeds the preset threshold D_b , the split is triggered. First, the shared vertices of the two adjacent triangular grids, V_1 and V_2 , are copied, meaning that their copies V_1' and V_2' are generated at the same position as the original shared vertices. Next, other triangular grids that share vertices V_1 and V_2 are determined, and then, the copied vertices V_1' and V_2' are used to reconstruct a triangular grid whose number is closer to m (including the m th triangular grid); the triangular grid whose number is closer to n (including the n th triangular grid) is not

reconstructed. These two steps break the original connections between the grids and realize the split process. Then, with further optimization, the copied vertices might gradually separate from the original vertices, which separates adjacent triangular grids. The split process of the triangular mesh from the inside and the edge is shown in Fig. 3b.

The split allows the mesh to form new branches and holes, which increases the deforming flexibility of the mesh and enables the mesh to change its topology by itself.

2.3.3 Refinement

In topology optimization, triangular meshes are deformed to optimize the overall layout. However, deformation sometimes leads to overlapping of the triangles. In such cases, coverage of the intersected triangle has little or no effect on the overall layout. Furthermore, the intersected triangle is locked up inside the geometry throughout the optimization process; therefore, it is not used in any subsequent steps. Because of these problems, the optimization algorithm is unable to work properly.

V_{nt} denotes the volume of the n th triangular grid, which can be calculated by Heron's formula as

$$p = \frac{1}{2}(l_{nt1} + l_{nt2} + l_{nt3}) \quad (8)$$

and

$$V_{nt} = t\sqrt{p(p-l_{nt1})(p-l_{nt2})(p-l_{nt3})} \quad (9)$$

where l_{nt} is the side length of the triangular grid. Because the vertex coordinates $p(x, y)$ of the triangular grid are known variables, the side length can be easily obtained. The term t is the thickness of the component, and $\bar{V}$ is the maximum available high thermal conductivity material.

The topology optimization is solved with a Eulerian description and fixed finite element mesh on the design domain Ω_D as shown in Fig. 5.

The corresponding problem formulation can be expressed in terms of the finite element mesh node value $\phi(\mathbf{c})$ as

$$\begin{aligned} &\text{Find } \mathbf{c} = (\mathbf{c}_1, \dots, \mathbf{c}_{np})^T \\ &\text{Minimize } S = S(\mathbf{c}, T) \\ &\text{s.t.} \\ &\mathbf{c} \in \mathbf{U}_c \\ &\int_{\Omega_D} H(\phi) \gamma \delta_{ij} T_i \bar{T}_j d\Omega \\ &= \int_{\Omega_D} H(\phi) Q \bar{T} d\Omega + \int_{\Omega_D} \delta(\phi) |\nabla \phi| \tilde{g} \bar{T} d\Omega - \int_{\Omega_D} \delta(\phi) |\nabla \phi| h T \bar{T} d\Omega \\ &\quad \forall \bar{T} \in \bar{Y} \\ &T = \bar{T}, \text{ on } \Gamma_T \\ &V(\mathbf{c}) \leq \bar{V} \end{aligned} \quad (10)$$

where $H(\phi)$ and $\delta(\phi)$ are the Heaviside and Dirac delta functions, respectively, defined by

$$H(\phi) = \frac{1}{\pi} \arctan[(\phi-1) \cdot f] + 0.5 \quad (11)$$

and

$$\delta(\phi) = \frac{d}{d\phi} H(\phi) \quad (12)$$

where f is a very large constant value; in this work $f=1000$.

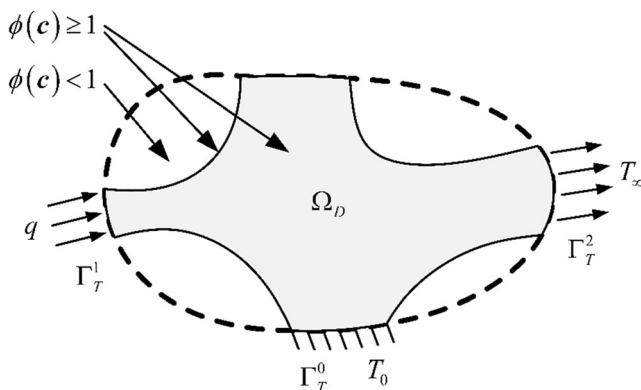


Fig. 5 Design domains and topology description function

It can be seen from Fig. 6 that the Heaviside function is continuous and differentiable, although most FEM nodes have values close to 0 or 1. It does not mean that the value of the FEM node cannot be a value between 0 and 1 (such as 0.5), although its probability is small. Because the phi function and the Heaviside function are continuous and differentiable, the stiffness factor function is also continuous and differentiable.

The $\tilde{h}$ and $\tilde{g}$ are defined as

$$\tilde{h} = \begin{cases} 0 & \phi \in \Gamma_T^0 \\ 0 & \phi \in \Gamma_T^1 \\ h & \phi \in \Gamma_T^2 \end{cases} \quad \text{and} \quad \tilde{g} = \begin{cases} 0 & \phi \in \Gamma_T^0 \\ q & \phi \in \Gamma_T^1 \\ hT_\infty & \phi \in \Gamma_T^2 \end{cases} \quad (13)$$

The procedure to obtain $\phi(\mathbf{c})$ is explained in the following discussion.

3 Numerical solution aspects

The TTM framework is described as a double-layer architecture in which we use a transformable triangular mesh as the component and a fixed background mesh with fine grids for finite element analysis. In this section, FEM layer-related techniques are analyzed, and parallel computing is discussed as well.

3.1 Finite element analysis

As depicted in Fig. 7, the component is explicitly described and divided into triangular grid, containing information such as vertex coordinates and vertex number. To control the upper layer component through the bottom layer finite element analysis, the upper component needs to be mapped with the background mesh, which is achieved by projection. Then, a value needs to be assigned to every node on the background mesh according to the projection relationship. If a node on the background

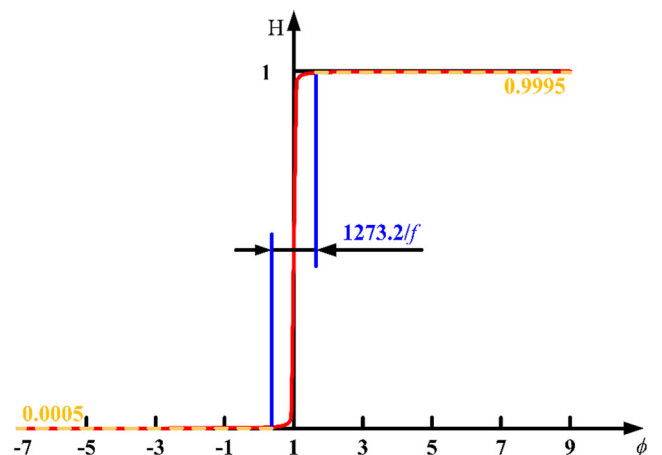
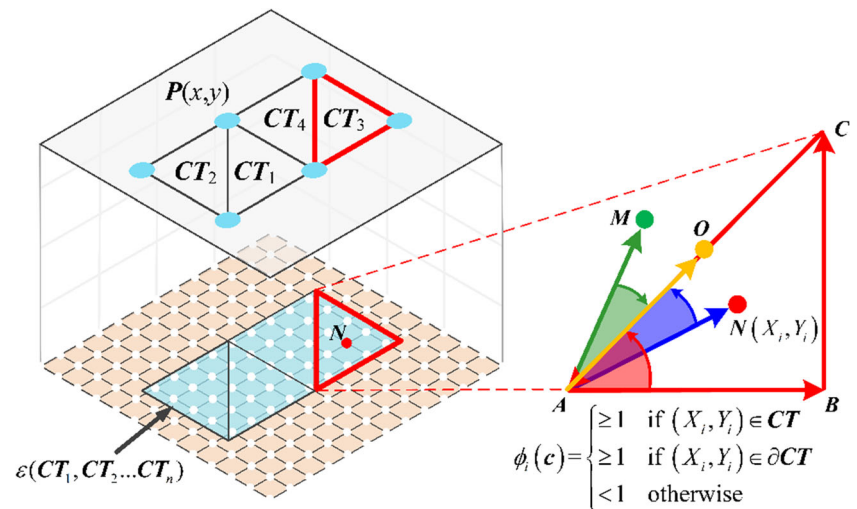


Fig. 6 Function image of the Heaviside function

Fig. 7 Projection scheme and determine if a node is inside the projection



mesh is in any of the component's triangular projections or on a triangular projection boundary, the node is set to greater than or equal to 1; otherwise, it is set to less than

1. The method to determine whether a background mesh node is inside the projection of one of the component's triangle is as follows: If a node is inside the triangle (node

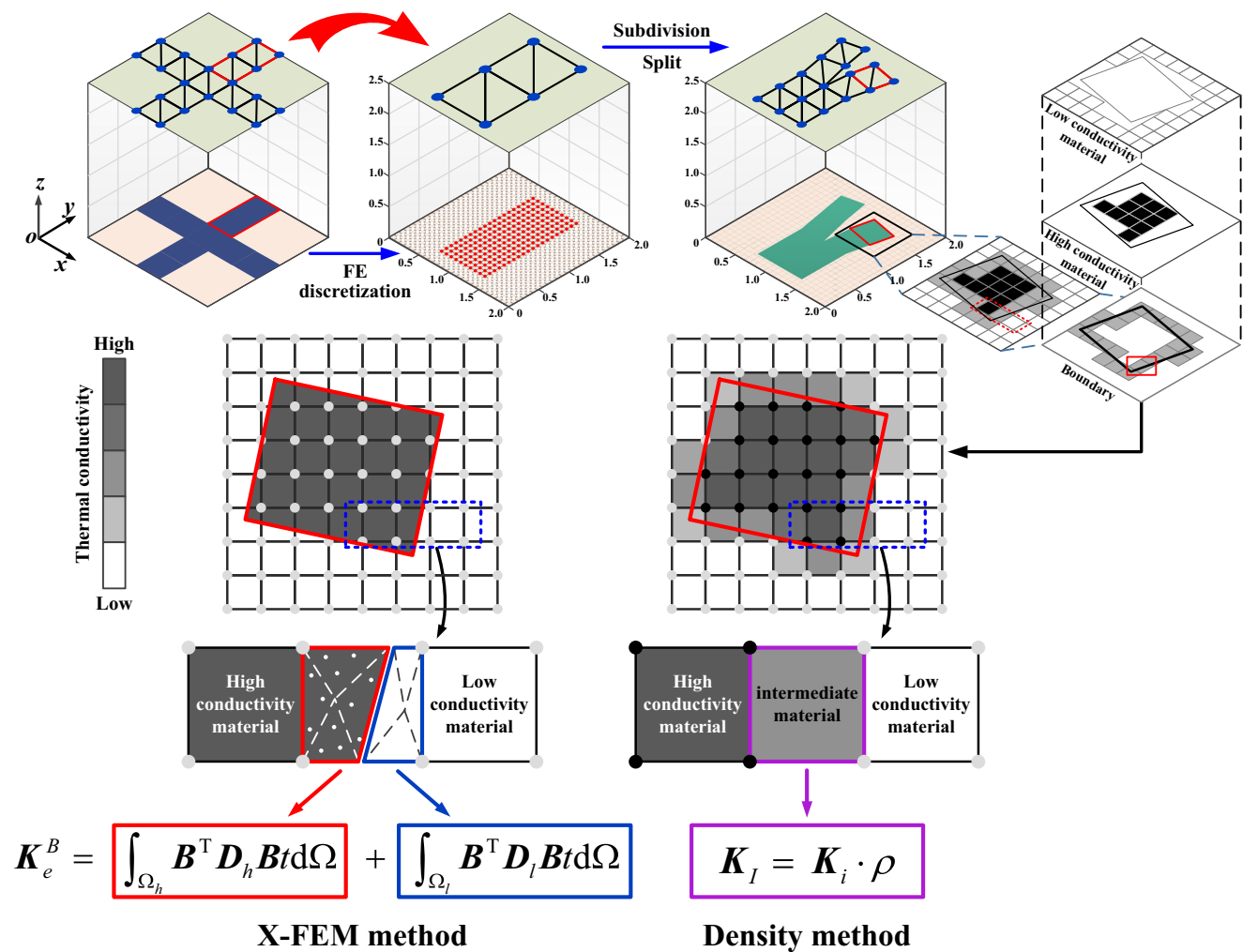


Fig. 8 TTM solution framework

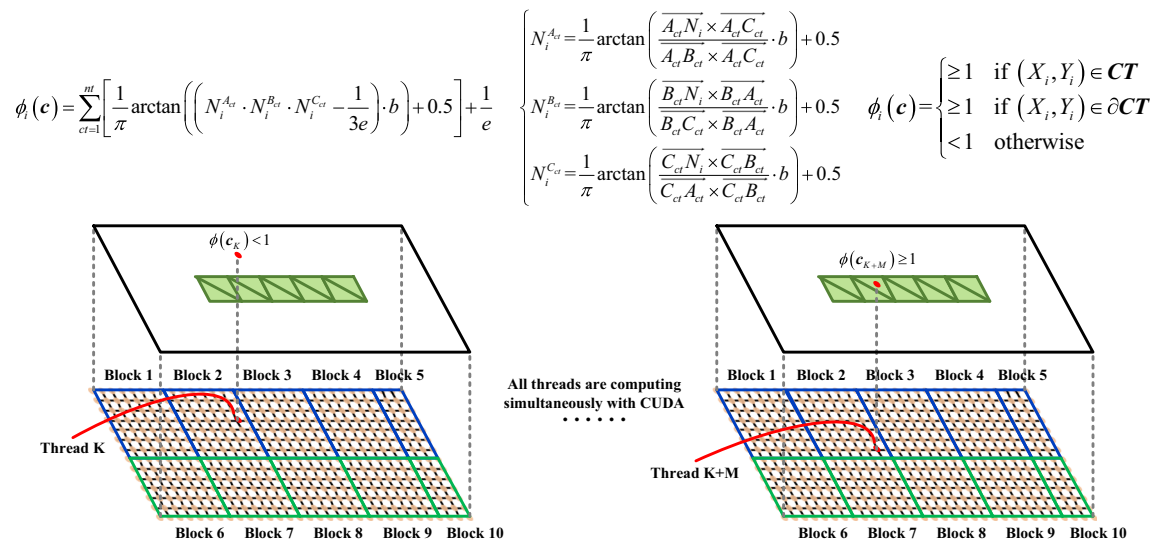


Fig. 9 Parallel computing implementation

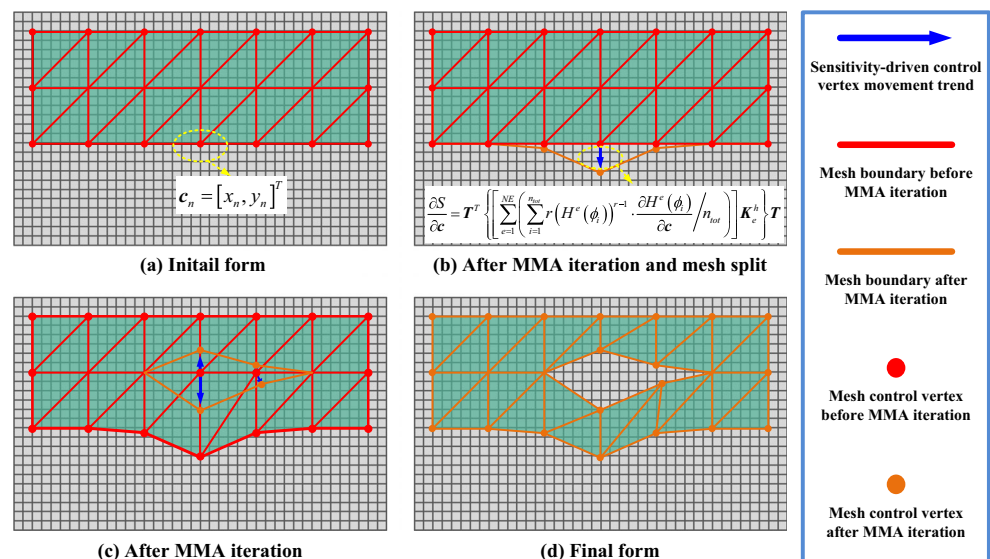
N in Fig. 7), when the triangle $\triangle ABC$ is viewed in the direction of AB , BC , CA (counterclockwise), node N is always to the left of these sides. For any node outside the triangle, this is not true (consider node M , for example). To implement this rule mathematically in the optimization process, a “side” condition can be realized by vector products. For example, if node N is inside the triangle, $\overrightarrow{AN} \times \overrightarrow{AC}$ and $\overrightarrow{AB} \times \overrightarrow{AC}$ should have the same sign, i.e., both are either positive or negative, and this should be true for all three sides. If node O is on the triangle boundary, $(\overrightarrow{AO} \times \overrightarrow{AC})(\overrightarrow{BO} \times \overrightarrow{BA})(\overrightarrow{CO} \times \overrightarrow{CB}) = 0$. In contrast, node M does not fulfill these conditions.

As shown in Fig. 7, $\phi_i(\mathbf{c})$ is the value of i th node in the design domain whose coordinates are (X_i, Y_i) , $\mathbf{CT}$ is the triangular grid of the component, $\partial \mathbf{CT}$ is the triangle boundary, ε denotes the component, and $P(x, y)$ is the vertex of the component.

In the present work, for example, the value of the i th node N_i can be written as

$$\phi_i(\mathbf{c}) = \sum_{ct=1}^{nt} \left[\frac{1}{\pi} \arctan \left(\left(N_i^{A_{ct}} \cdot N_i^{B_{ct}} \cdot N_i^{C_{ct}} - \frac{1}{3e} \right) \cdot b \right) + 0.5 \right] + \frac{1}{e} \quad (14)$$

Fig. 10 Process of sensitivity-driving component



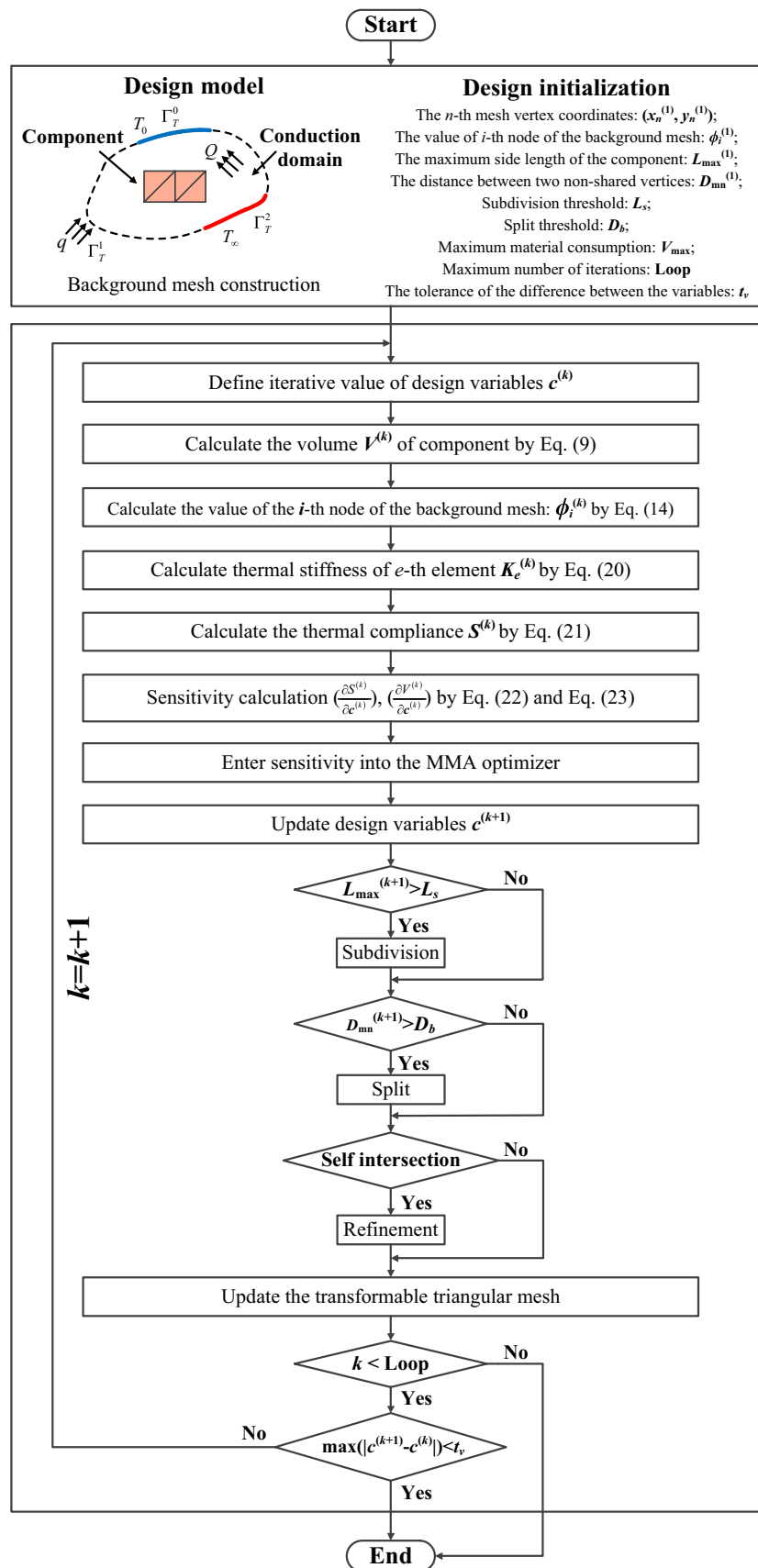


Fig. 11 Optimization flowchart

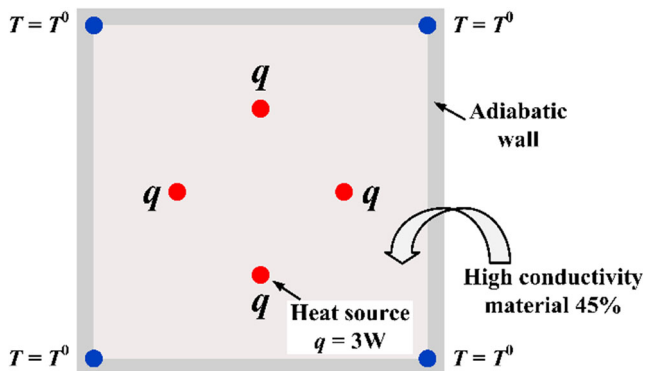
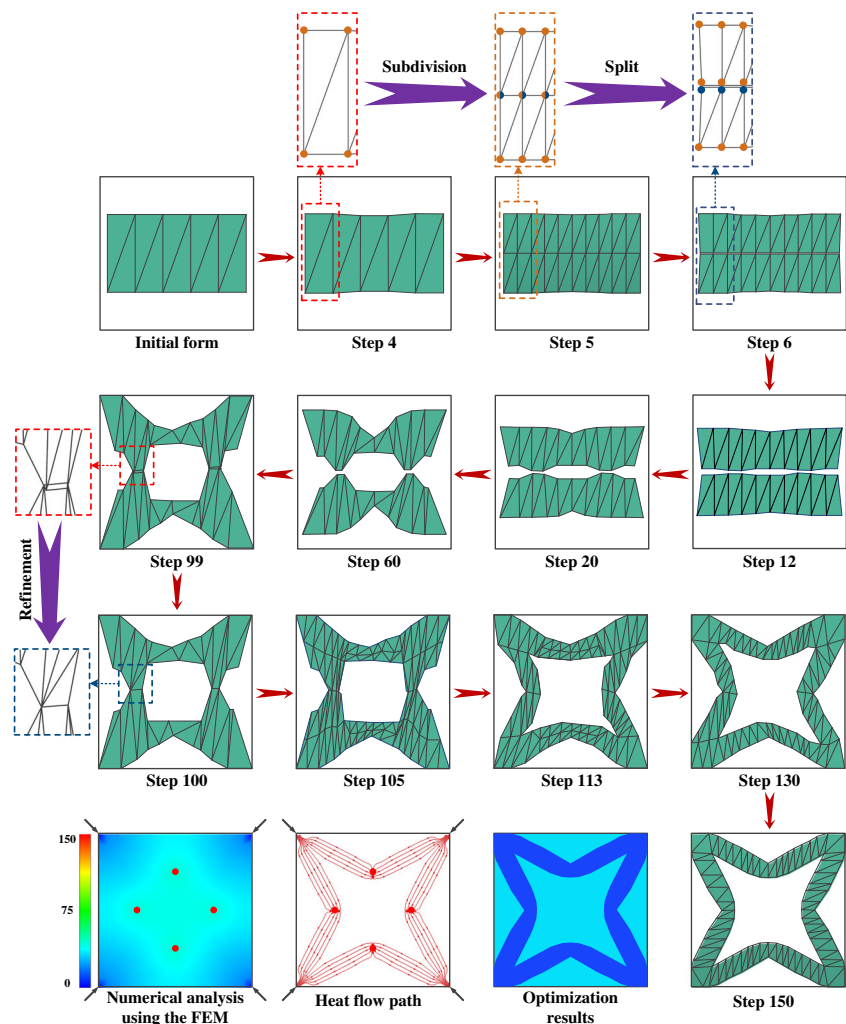


Fig. 12 Boundary condition of point source heat transfer problem

and

$$\begin{cases} N_i^{A_{ct}} = \frac{1}{\pi} \arctan \left(\frac{\overrightarrow{A_{ct}N_i} \times \overrightarrow{A_{ct}C_{ct}}}{\overrightarrow{A_{ct}B_{ct}} \times \overrightarrow{A_{ct}C_{ct}}} \cdot b \right) + 0.5 \\ N_i^{B_{ct}} = \frac{1}{\pi} \arctan \left(\frac{\overrightarrow{B_{ct}N_i} \times \overrightarrow{B_{ct}A_{ct}}}{\overrightarrow{B_{ct}C_{ct}} \times \overrightarrow{B_{ct}A_{ct}}} \cdot b \right) + 0.5 \\ N_i^{C_{ct}} = \frac{1}{\pi} \arctan \left(\frac{\overrightarrow{C_{ct}N_i} \times \overrightarrow{C_{ct}B_{ct}}}{\overrightarrow{C_{ct}A_{ct}} \times \overrightarrow{C_{ct}B_{ct}}} \cdot b \right) + 0.5 \end{cases} \quad (15)$$

Fig. 13 Optimization process of point source heat transfer problem



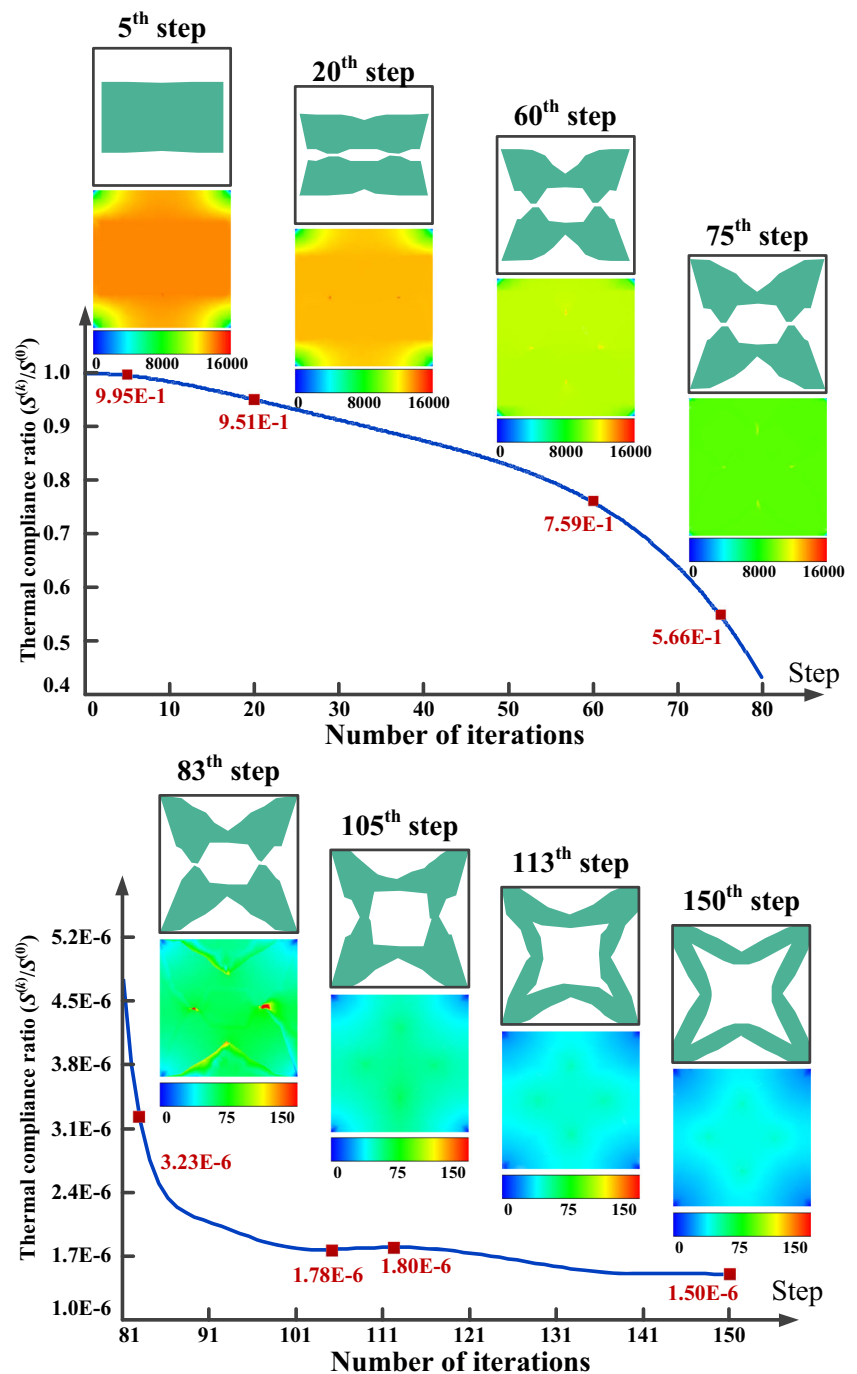
where $\overrightarrow{A_{ct}N_i} \times \overrightarrow{A_{ct}C_{ct}} = (X_{N_i} - x_{A_{ct}})(y_{C_{ct}} - y_{A_{ct}}) - (x_{C_{ct}} - x_{A_{ct}})(y_{N_i} - y_{A_{ct}})$ and $\overrightarrow{A_{ct}B_{ct}} \times \overrightarrow{A_{ct}C_{ct}} = (x_{B_{ct}} - x_{A_{ct}})(y_{C_{ct}} - y_{A_{ct}}) - (x_{C_{ct}} - x_{A_{ct}})(y_{B_{ct}} - y_{A_{ct}})$. The term nt is the number of triangular grids of the component, e is the Euler's number, and b is a large constant (e.g., $b = 1,000,000$).

Taking advantage of the above assignment, the structural attribute (existence or extinction) of any node can be determined from its value. Furthermore, it forms the exact geometrical description of the high thermal conductivity material component.

The projection and its processing are shown in Fig. 8.

The explicit geometrical description clearly describes the continuous boundary of the component. Thus, the boundary of the component may intersect with the background mesh, which will require remeshing. One solution that can solve this problem accurately is X-FEM (Elguedj et al. 2018), where the intersected elements consist of two parts: high thermal conductivity material and low thermal conductivity material. The thermal stiffness matrix can be calculated as

Fig. 14 Changing history of the objective function and the temperature distribution



$$\mathbf{K}_e^B = \mathbf{K}_h + \mathbf{K}_l \quad (16)$$

and

$$\begin{cases} \mathbf{K}_h = \int_{\Omega_h} \mathbf{B}^T \mathbf{D}_h \mathbf{B} t d\Omega \\ \mathbf{K}_l = \int_{\Omega_l} \mathbf{B}^T \mathbf{D}_l \mathbf{B} t d\Omega \end{cases} \quad (17)$$

where $\mathbf{K}_h$ and $\mathbf{K}_l$ are the contributions of the high thermal conductivity material and low thermal conductivity

material, respectively, to the entire thermal stiffness matrix, matrix $\mathbf{B}$ is the temperature differential matrix, matrices $\mathbf{D}_h$ and $\mathbf{D}_l$ are the thermal conductivity matrices for the high and low thermal conductivity materials, respectively, and t is the thickness of the element.

In the X-FEM scheme, both high and low thermal conductivity materials can be further divided into numerous small triangles. The contribution for the entire element of each small triangle can be easily obtained from the Gaussian integral. The

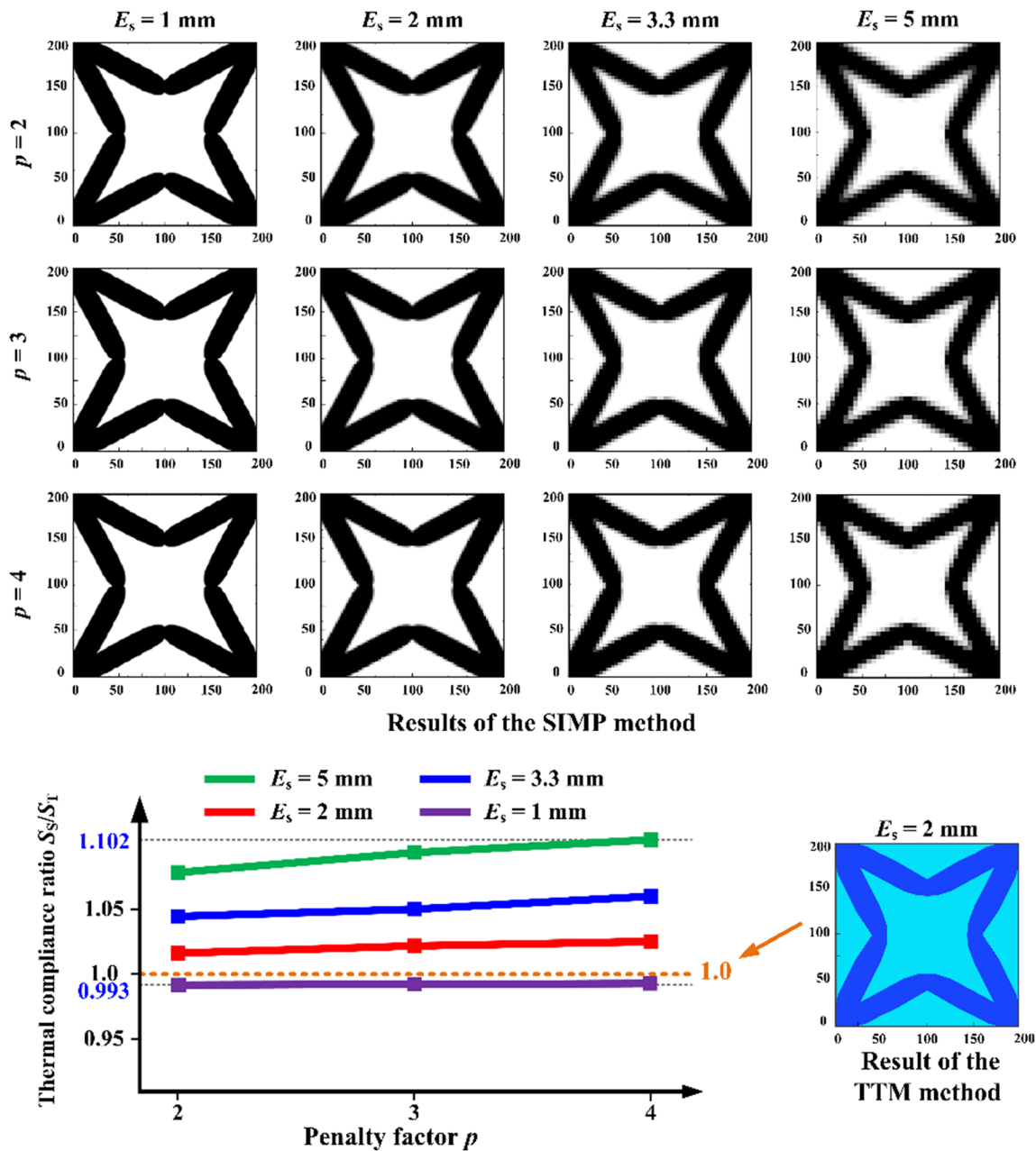


Fig. 15 Comparison of the results obtained by the TTM method and SIMP method

entire thermal stiffness matrix of the intersected elements can be calculated by the summation of the above Gaussian integrals.

Another solution for the boundary projection problem is the density method, which utilizes an intermediate material known as the ersatz material (Xu et al. 2004) to describe the thermal conductivity of each element. The element thermal stiffness matrix is written as

$$\mathbf{K}_I = \mathbf{K}_i \cdot \rho \quad (18)$$

where $\mathbf{K}_I$ is the thermal stiffness matrix of the intersected element, $\mathbf{K}_i$ is the thermal stiffness matrix of the integrated

element, and ρ is the pseudo density of the element valued by the proportion of the high thermal conductivity material.

The advantage of the X-FEM is its ability to accurately identify the distribution of the material. However, to enhance the computational efficiency, the density method is used for FEM analysis in the present work. In the numerical calculation of this method, the accuracy can also be preserved if an appropriate mesh size is chosen.

The thermal conductivity of the intermediate material is formulated as

$$E_{act} = E \times \frac{\sum_{i=1}^{n_{tot}} (H(\phi_i))^r}{n_{tot}} \quad (19)$$

Table 1 Comparison of computation time between the TTM method and SIMP method

Element size (mm)	Penalty factor	SIMP Time	TTM method Time
$E_s = 1.0$	$p = 2$	537.42 s	465.62 s
	$p = 3$	529.90 s	
	$p = 4$	540.11 s	
$E_s = 2.0$	$p = 2$	297.03 s	240.23 s
	$p = 3$	309.34 s	
	$p = 4$	318.69 s	
$E_s = 3.3$	$p = 2$	145.25 s	142.72 s
	$p = 3$	211.92 s	
	$p = 4$	257.53 s	
$E_s = 5.0$	$p = 2$	100.42 s	110.65 s
	$p = 3$	165.97 s	
	$p = 4$	206.01 s	

where E_{act} is the actual thermal conductivity of an element, E is the thermal conductivity of the original material, n_{tot} is the total number of nodes embedded in a single element, $r = 2$ in this work, and $H(\phi_i)$ is defined in Eq. (11).

Based on this treatment, it is easy to obtain the thermal stiffness of the e th element in the FEM framework as

$$\mathbf{K}_e = \frac{\sum_{i=1}^{n_{tot}} (H(\phi_i))^r}{n_{tot}} \mathbf{K}_e^h \quad (20)$$

where $\mathbf{K}_e^h$ is the thermal conductivity matrix of the element with the high thermal conductivity material.

The thermal compliance S can be written as

$$\begin{aligned} S &= \mathbf{T}^T \mathbf{K} \mathbf{T} \\ &= \mathbf{T}^T \left\{ \sum_{e=1}^{NE} \left[\sum_{i=1}^{n_{tot}} (H^e(\phi_i))^r / n_{tot} \right] \mathbf{K}_e^h \right\} \mathbf{T} \end{aligned} \quad (21)$$

where K is the total thermal stiffness matrix, T is the node temperature matrix, NE is the number of background mesh elements, and $H^e(\phi_i)$ is the Heaviside function value of the i th node of the e th element whose coordinates are (X_i^e, Y_i^e) .

3.2 Design sensitivity

The sensitivity of thermal compliance S with respect to variable $\mathbf{c}$ can be written as

$$\frac{\partial S}{\partial \mathbf{c}} = \mathbf{T}^T \left\{ \left[\sum_{e=1}^{NE} \left(\sum_{i=1}^{n_{tot}} r (H^e(\phi_i))^{r-1} \cdot \frac{\partial H^e(\phi_i)}{\partial \mathbf{c}} / n_{tot} \right) \right] \mathbf{K}_e^h \right\} \mathbf{T} \quad (22)$$

Because $(x_{ct}, y_{ct}) \in \mathbf{c}$ in Eq. (15), $\frac{\partial \phi_i(\mathbf{c})}{\partial \mathbf{c}}$ can be easily obtained. The sensitivity of thermal compliance S with respect to variable $\mathbf{c}$ can also be obtained.

In addition, it also has

$$\frac{\partial V}{\partial \mathbf{c}} = \sum_{e=1}^{NE} \left(\sum_{i=1}^{n_{tot}} \frac{\partial H^e(\phi_i)}{\partial \mathbf{c}} / n_{tot} \right) \quad (23)$$

In the present work, the method of moving asymptotes (MMA) optimizer is used to update the parameters in the optimization iterations.

3.3 The parallel computing implementation

In the topology optimization process, both the subdivision operation and the split operation of the TTM increase the number of vertices, which gradually increases the computational complexity during the iterations. In addition to some computational resources spent in the MMA solver, computational resources are mainly used in the projection step between the TTM and the background mesh during each iteration.

The projection between the TTM and the background mesh is achieved by traversing each background mesh node, judging whether these nodes are inside or on the boundaries of any triangular grid in the TTM projection. One of the characteristics of this kind of processing is that the calculation of any one node does not affect other nodes, which means that any iteration in the traversal does not interfere with others. The Nvidia CUDA GPU computing technology is ideal for parallel traversals under this characterization.

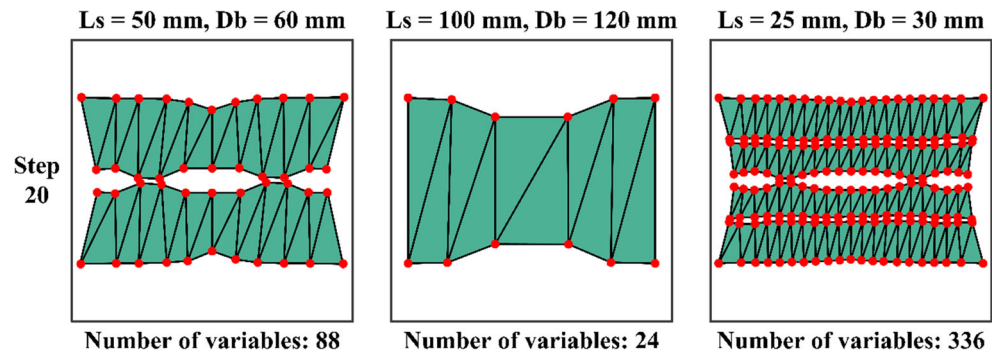
The graphics card consists of several grids, and each grid can be further divided into blocks. Hundreds of threads can run simultaneously in each block. There are three main operations in the CUDA parallel framework. First, the number of grids and blocks required for traversal needs to be checked. This is determined by the total number of nodes to be checked and the specifications of the graphics card (i.e., the threads per block, blocks per grid, and number of grids). Second, the node data is copied from the SD RAM to the high-speed graphics shared memory of the specific block. Finally, the TTM geometry information is copied to all blocks so that it can be acquired by each thread in the block in parallel.

The computation in each thread is the traversal of each node for each triangular mesh in the TTM projection. To save computational cost, once the calculation shows that the node is within the triangle mesh or boundary during the traversal of a node, the traversal stops immediately and returns the result. A contiguous area is allocated from the shared memory so that each thread can write the result to its specific address. The results are passed back to the SD RAM from the graphics shared memory for further steps after all calculations have been completed.

A schematic diagram of the implementation of parallel computing is shown in Fig. 9.

This operation reduces the computational cost of the topology optimization process, and a comparison of the

Fig. 16 Effect of L_s and D_b thresholds on optimization



computational cost between the two methods is shown in Sect. 4.1 later.

3.4 Topology optimization process

The entire topology optimization process begins with the initialization for both TTM and the background mesh, and it is followed by the iterative process, which starts from the projection between TTM and the background mesh. The properties of each node in the background mesh are edited according to the projection relationship so that the thermal compliance and its sensitivity can be further calculated. The component volume and volume sensitivity can be calculated without projection because of TTM's explicit geometric description. After that, the sensitivities are used as input to the MMA optimization, whereas new variables (control vertex coordinates of the TTM) serve as output. Then, mesh operations are used to update the geometry of the TTM and the connection relationship in preparation for the next iteration. Figure 10 shows how the sensitivity drives the component.

At the end of an iterative process, two termination conditions of the iteration should be taken into consideration. One is that the difference between the variables for the two iterations is smaller than a predetermined tolerance and the other is the number of iterations. The algorithm stops if any of the

conditions are met; otherwise, it proceeds to the next iteration. The entire topology optimization process is shown in Fig. 11.

Unlike early methods, which relied on multiple components to form the optimal structure, this optimization method fully stimulates the deformation and topology-changing ability of a single component, and thus, an optimal structure with only one component can be achieved.

4 Case study

In this section, two cases are presented to demonstrate the effectiveness of the proposed approach.

4.1 Case study 1

Figure 12 depicts a design domain of a 200×200 mm square plate with four adiabatic sides. Four vertex points are set as heat sinks with a constant temperature $T = T^0 = 273$ K. As depicted in the figure, four heat source points are placed with heat generated at a rate of $q = 3$ W. The original design domain is filled with a low thermal conductivity material, $E_b = 0.3$ W/(m K), and the optimized heat-conducting path is made of a high thermal conductivity material, $E = 141$ W/(m K), where $E/E_b = 470$. The thickness of the plate is $h = 2$ mm, and the volume ratio of the high thermal conductivity material is set as 45%. The design domain is discretized into a 100×100 mesh.

Figure 13 presents the whole optimization process. Initially, the component is just a simple rectangular block disconnected from the heat sink points. As the optimization proceeds, the component shape changes dramatically, gradually approaching the heat sink points and the heat source points. The mesh is densified at step 5, and the split action is triggered at step 6. This process is completed at step 12, with the mesh fully divided into two parts. Then, the mesh starts to connect with the heat source points and the heat sink points. It is obvious that, at step 100, a refinement is executed to deal with the overlapping of triangular grids. Finally, after the 150th step in the iteration, the optimal structure is obtained. It is a

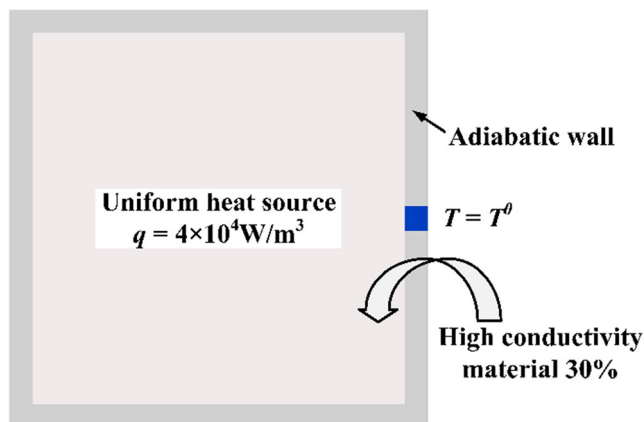
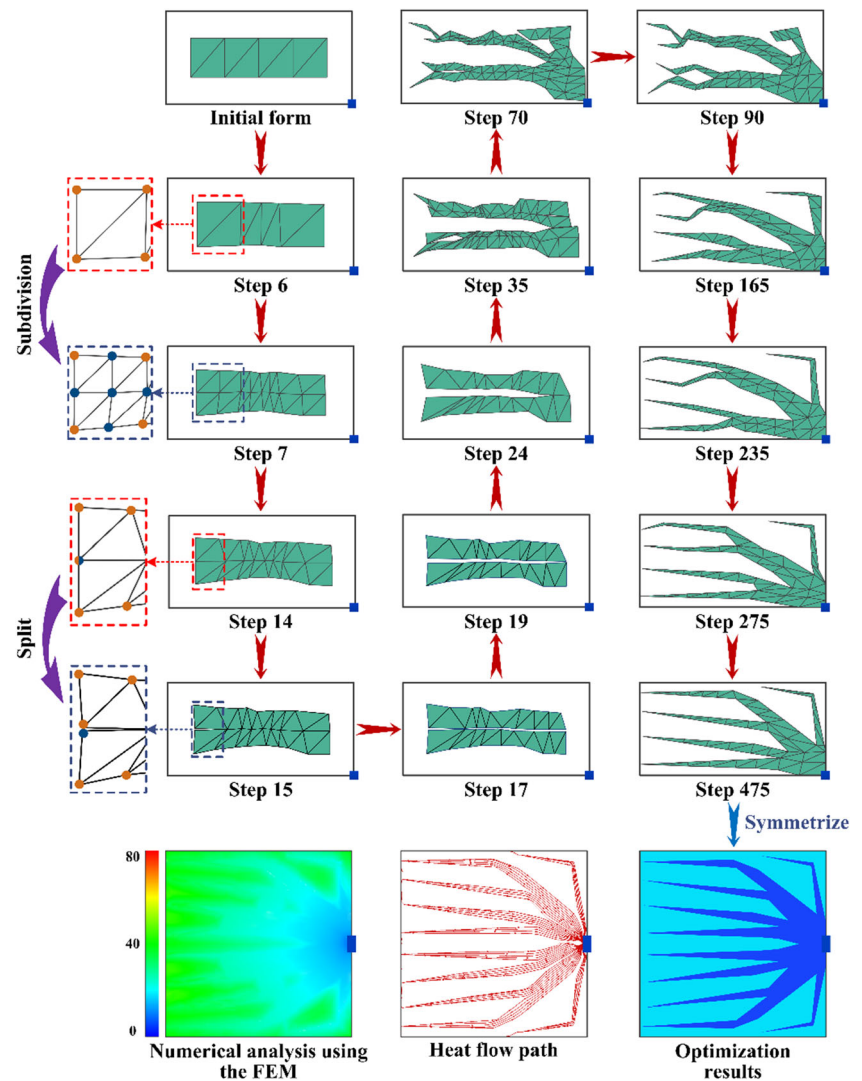


Fig. 17 Boundary condition of surface source heat transfer problem

Fig. 18 Optimization process of surface source heat transfer problem



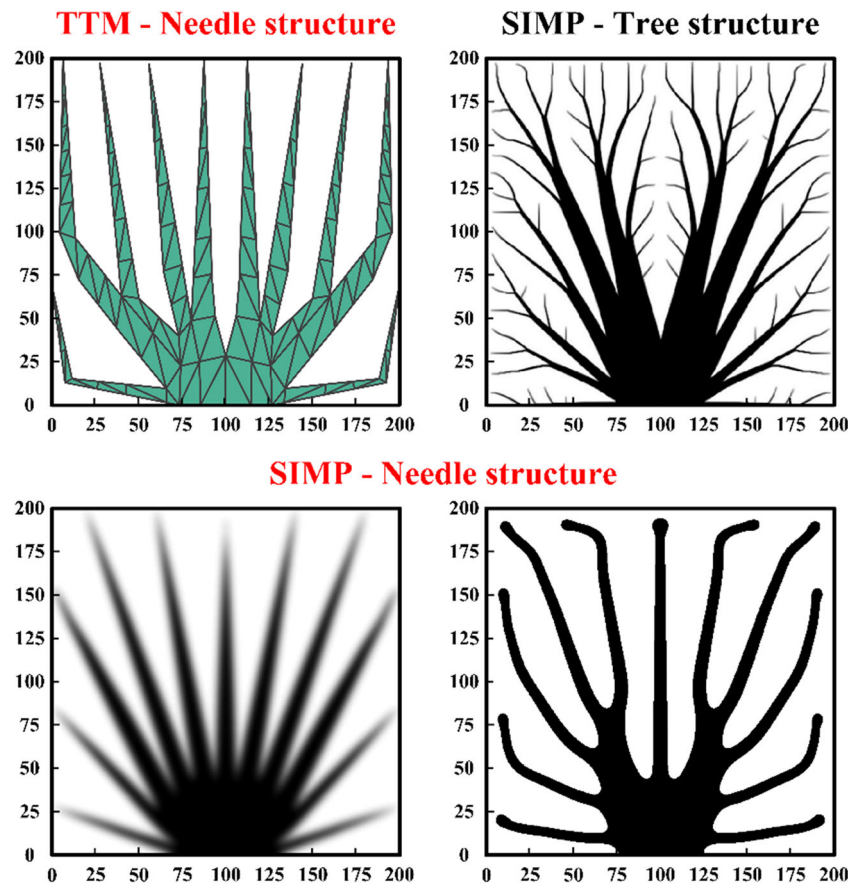
symmetric structure where each heat sink connects with two heat sources by two paths directly, which is also the classic optimal result for this heat conduction problem. Throughout the optimization process, the deformation, subdivision, split, and refinement actions are activated multiple times. What we can see from the heat flow chart is that the heat generated by the heat sources is transferred directly to the heat sinks via the heat paths. A highly uniform temperature distribution is also achieved.

The evolutionary history of the objective function and the temperature distribution is shown in Fig. 14. A significant decrease in the objective happens in the first 80 steps; the rest of the optimization continues this trend with minor changes. When the heat paths first connect the heat sources with the heat sinks, which happens at step 81, the performance of the entire structure is greatly improved. The result proves that the presented method performs well in achieving an optimal structure in heat path optimization problems with multiple heat sources and heat sinks.

Figure 15 presents the optimization results of the SIMP method with different mesh sizes E_S and penalty factors p and the results obtained by the proposed TTM method. The comparison was implemented on a workstation with an Intel (R) Xeon (R) E5-2609 v4 processor, 32GB RAM, and a 64-bit operating system. Illustrated by the case of $E_S = 2$ mm, a comparison of computation time between the TTM method and the SIMP method is shown in Table 1. Compared with the SIMP method under different values of p , the computation time for the TTM method (240.23 s) decreased by 19.1% ($p = 2$), 22.3% ($p = 3$), and 24.6% ($p = 4$). This proves that the TTM method is capable of obtaining optimal structures with only one component with high computational efficiency.

Remark 1 In the present work, before the first split, the value of L_s is 25% of the shortest side length of the design domain, and the value of D_b is 30% of the shortest side length of the design domain. After each split, the

Fig. 19 Needle structure obtained by TTM method and SIMP method



thresholds of L_s and D_b are reduced 10%, which is the appropriate solution after using multiple sets of case trials, applicable to the cases that have been applied at this stage. If the values of L_s and D_b are too small, the transformable triangular mesh will be over-subdivided and over-split, which will increase the number of control vertices in the transformable triangular mesh, causing the variables to increase, resulting in increased computational cost. If the values of L_s and D_b are too large, the mesh will not subdivide and split, which will reduce the flexibility of the transformable triangular mesh. These conditions are shown in Fig. 16.

4.2 Case study 2

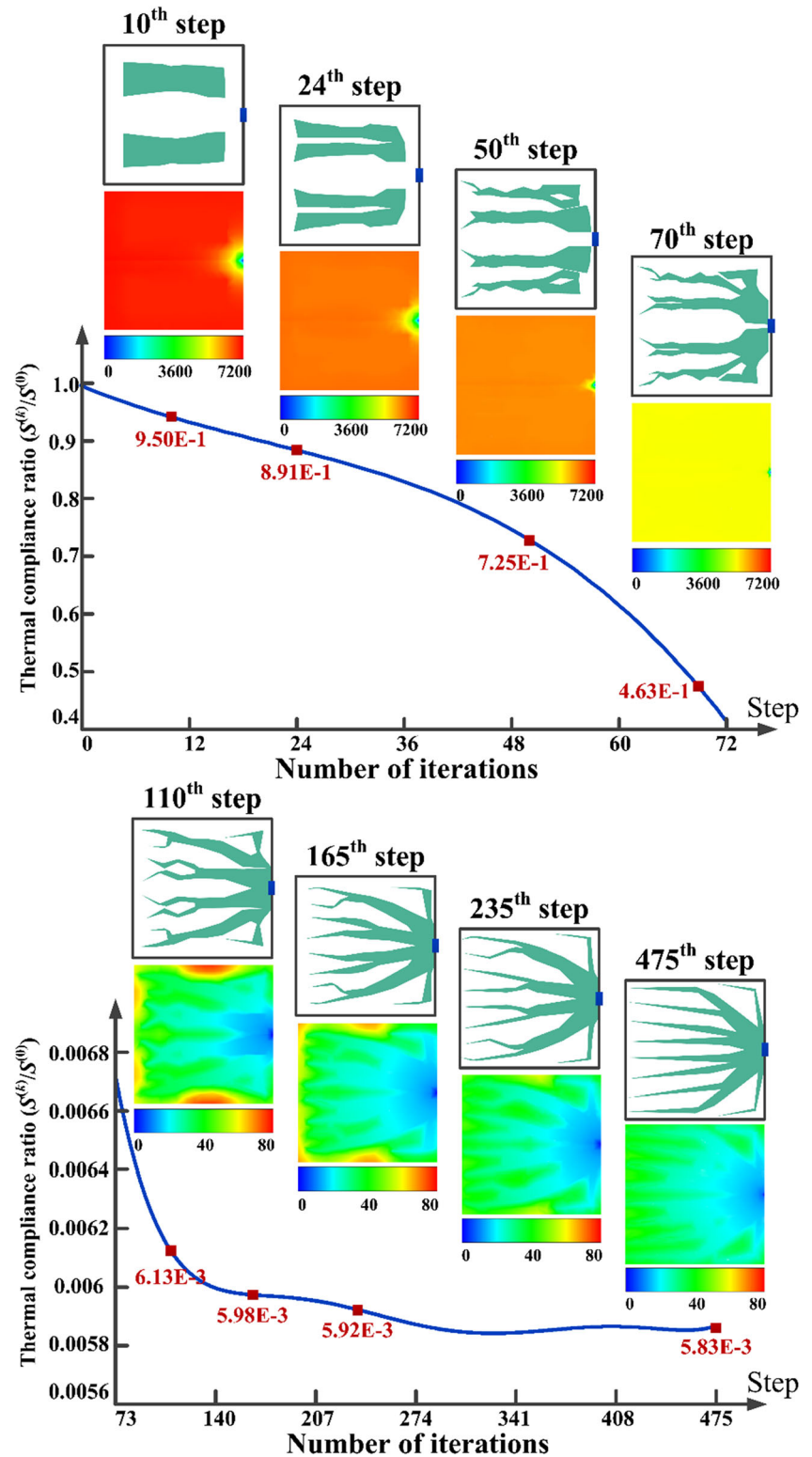
Figure 17 shows a design domain of a 200 mm $\times$ 200 mm square plate with four adiabatic sides. The midpoint of one of the sides is set as the heat sink with constant temperature $T = T^0 = 273$ K. Heat is generated uniformly across the whole design domain at a rate of $q = 4 \times 10^4$ W/m³. The original design domain is filled with a low thermal conductivity material $E_b = 0.3$ W/(m K), and the optimized heat-conducting path is made of a high thermal

conductivity material $E = 141$ W/(m K), where $E/E_b = 470$. The thickness of the plate $h = 2$ mm, and the volume ratio of the high thermal conductivity material is set as 30%. The design domain is discretized using a 100×100 mesh.

The optimization process is shown in Fig. 18. In view of the symmetries of loading condition and the final result, the optimization is carried out on only half of the design domain. The initial component still uses a simple rectangular block. As we can see, subdivision and split also happen at step 7 and step 15, respectively. After 475 steps of iteration, the optimal structure is obtained. In this case, the result is a tree-shaped structure with the “root” located at the heat sink and several “branches” extended throughout the design domain. This structure makes sense because the branches can collect the heat generated on the domain and transfer it to the root—heat sink, as shown in the heat flow path. Highly uniform temperature distribution on the plate is also achieved in this result.

As shown in Fig. 19, the optimal structure obtained by the TTM method in the surface source heat transfer problem is a needle-like structure, which is similar to the heat conduction

Fig. 20 Changing history of the objective function and temperature distribution



structure obtained in the work of Sigmund in 2018 (Yan et al. 2018). This SIMP-based research proves the non-optimality of the thermal tree structure and the excellent thermal conductivity of the needle structure in the surface source heat transfer problem.

Figure 20 depicts a history of the development for the objective function and the temperature distribution. As in the previous case study, the objective decreases in two stages with different decreasing rates, and a highly uniform temperature distribution is achieved on the optimal

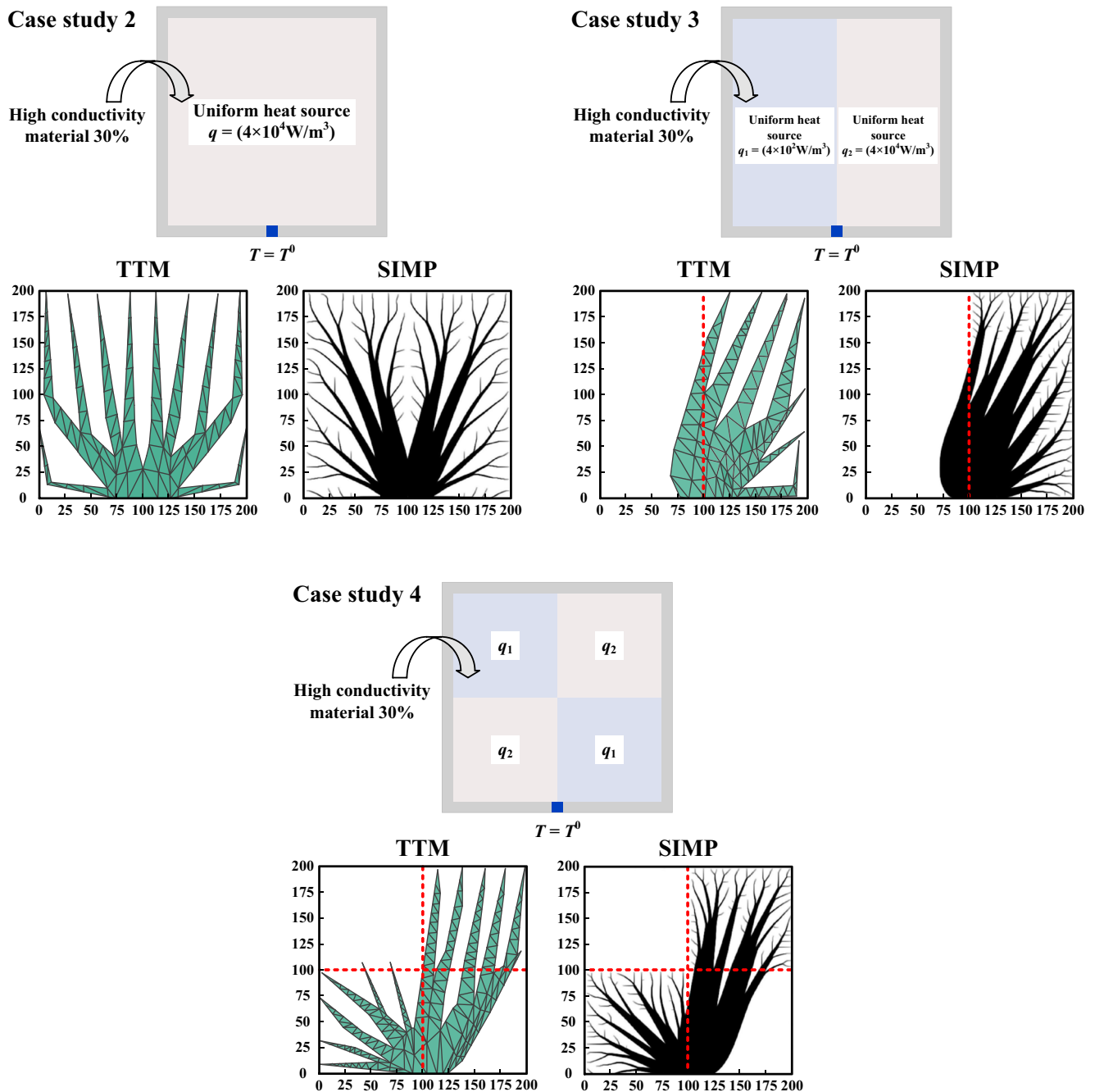


Fig. 21 Boundary conditions and corresponding results obtained with the TTM method and the SIMP method

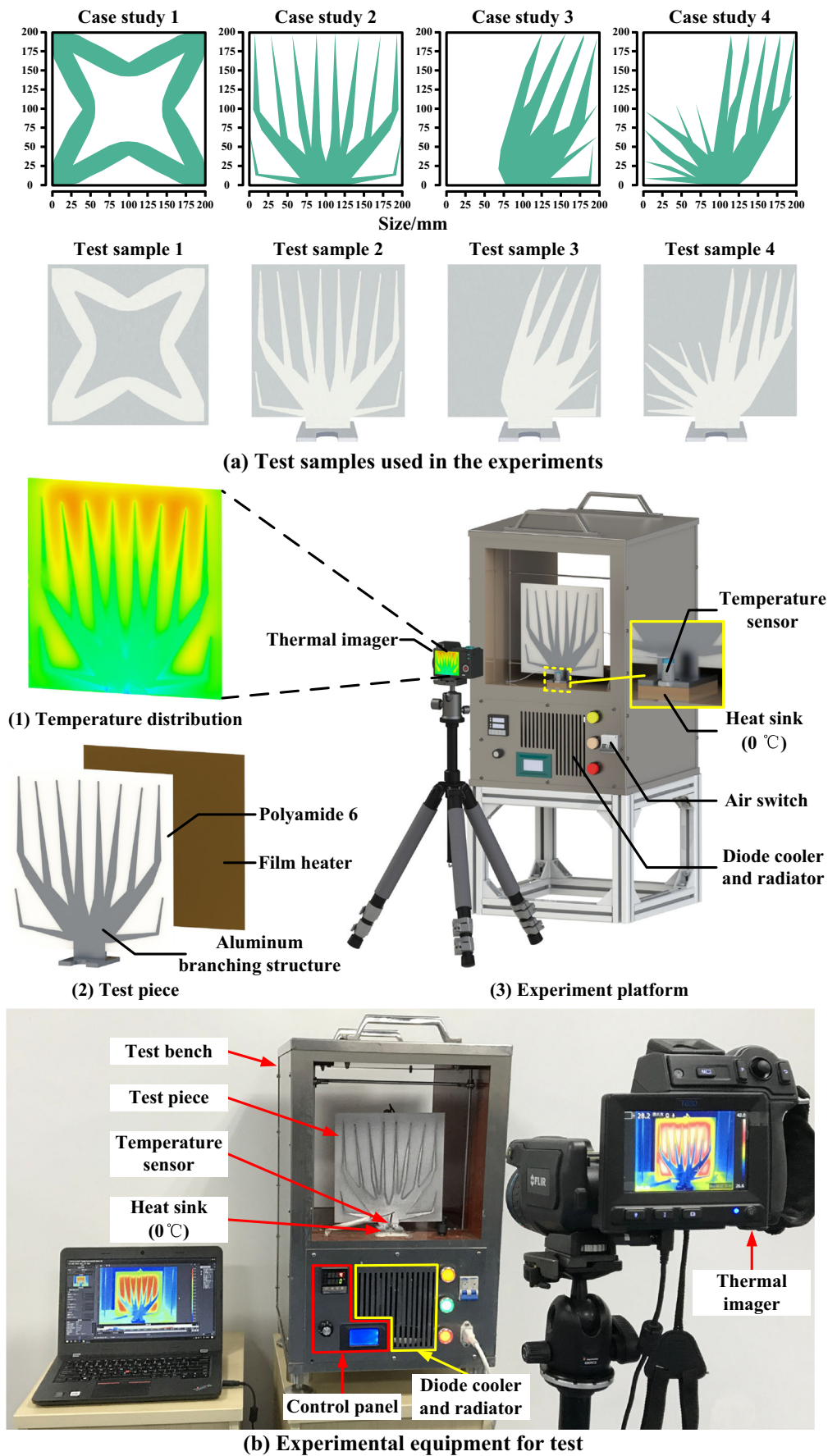
structure. The result proves that the presented method performs well in achieving optimal structures in heat path optimization problems with a uniform area heat source.

In addition, to further verify the robustness of this method, two different heating conditions are applied on this case study, as depicted in Fig. 21. The design domain is divided into different parts with two heat generation rates, $q_1 = 4 \times 10^2 \text{ W/m}^3$ and $q_2 = 4 \times 10^4 \text{ W/m}^3$. Comparison of the results obtained by the SIMP method demonstrates that the two structures are similar. This further proves the effectiveness and robustness of the proposed TTM method.

4.3 Experimental test

To verify the correctness of the simulation results, experimental tests were conducted as well. We selected 4032 aluminum alloy [$k = 138\text{--}154 \text{ W/(m K)}$] and Polyamide 6 [$k = 0.26\text{--}0.35 \text{ W/(m K)}$] as the high and low conductivity materials, respectively, so that the conductivity ratio equaled 470. The test samples, which are depicted in Fig. 22a, were prepared by wire-cut electrical discharge machining (WEDM). The experimental settings are illustrated in Fig. 22b. The thermal load was applied to the test sample by a film heater, and a

Fig. 22 Thermal experiment of different test cases



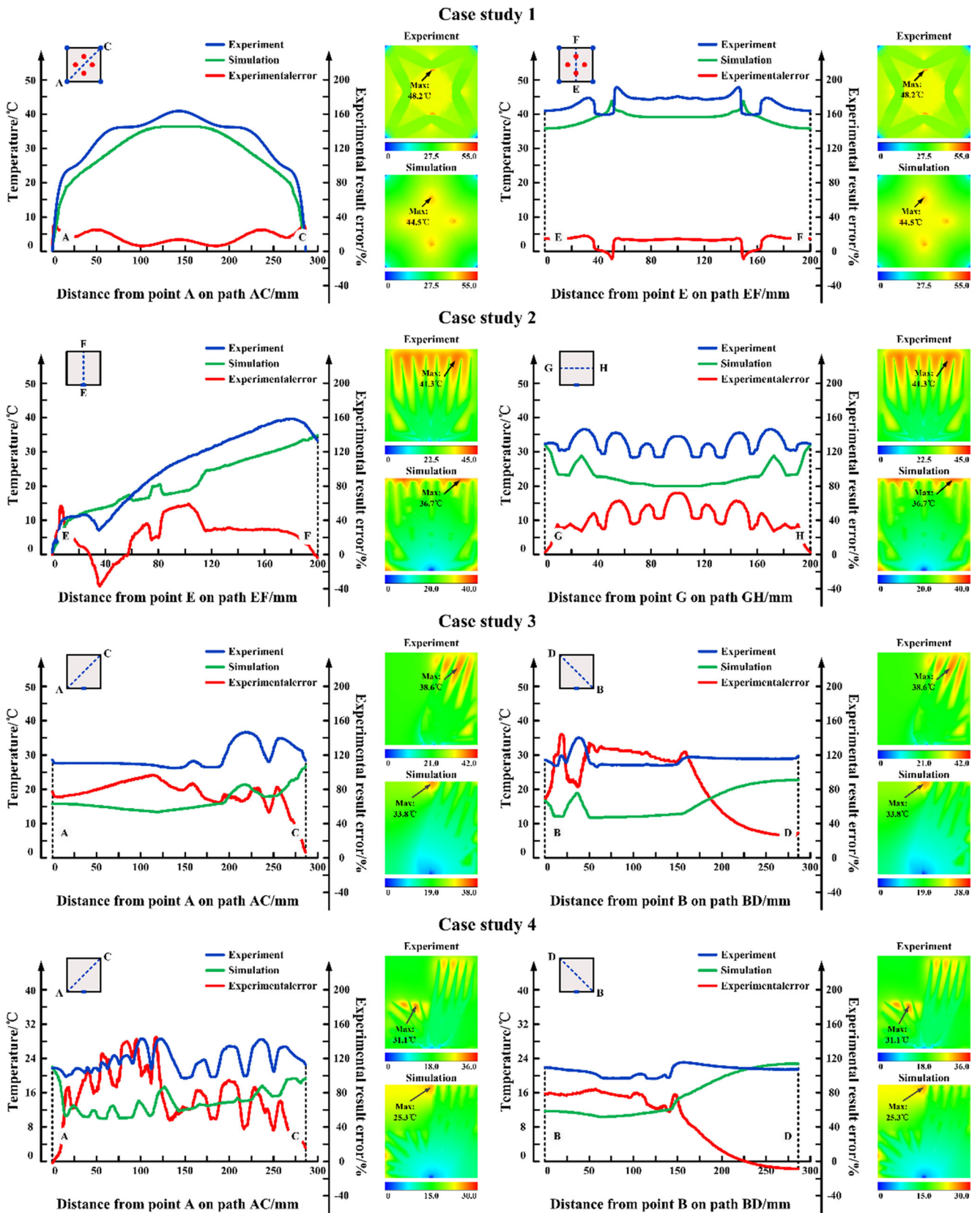


Fig. 23 Comparison of experimental and numerical results for different cases

semiconductor cooling chip (TEC1-12710) was used to keep the temperature of the heat sink constant.

An infrared thermal imager (SC7300M) was applied to test the temperature distribution. A comparison between the simulation results and the experimental results is presented in Fig. 23.

For case study 1, paths AC and EF were selected for measurement because the structure was centrally symmetric. It can be seen from the temperature curve measured along path AC that the experimental results are generally higher than those obtained by the simulation, but the trends of the curves are similar. However, for the path EF measurement, the experimental temperature curve is a little different from the simulated temperature curve, which may have been caused by the poor performance of the thermal grease between the high and low conductivity materials. Therefore, a much higher temperature was obtained for the low conductivity material than for the high conductivity material in the same vicinity. For case study 2, its structure was axisymmetric. In this work, paths EF and GH were selected for measurement. It can be observed from the results measured along path EF that the experimental curve decreases in the interval between 22 mm and 38 mm and in the interval between 180 mm and 200 mm, which may have been caused by uneven heating of the film heater. Along path EF, the difference between the shape of the experimental temperature curve and that of the simulated temperature curve may be attributed to poor thermal conduction between the high and low conductivity materials. For case study 3 and case study 4, the structures were asymmetrical, and paths AC and BD were selected for measurement in this work. Although the temperature curves obtained by experiment in cases 3 and 4 are similar to their respective simulated temperature curves, the position of the highest temperature in the cloud diagram is different, which may also have been due to poor thermal conduction between the high and low conductivity materials.

In summary, although the temperature values of the experimental results are generally higher than those of the simulation results, shapes of the temperature curves are similar, which proves the correctness of the numerical simulation. The temperature distribution is uniform and the heat dissipation is efficient in each case.

5 Conclusion

In this study, an explicit topology optimization method based on a TTM is proposed. This approach can generate optimal structural topologies with only a single component, which overcomes the dependence of traditional explicit approaches on the initial configurations. This breakthrough is achieved as the component based on the idea of geometric morphing can obtain high deformability through related mesh

transformation operations; therefore, this method is characterized by extremely high topology-changing capabilities. Based on this, we numerically describe the mathematical model and set up cases to verify the effectiveness and applicability of the proposed method. The results demonstrate that the method is not only capable of obtaining optimal structures with only one component and explicitly describing the structure, but it is also computationally efficient and significantly robust under different loading conditions. In future work, this method will be extended to 3D, which will require the creation of a highly flexible tetrahedron as the component. The mesh processing techniques, such as mesh split, can be realized in tetrahedrons.

Replication of results The results presented in the paper can be reproduced. We do not want to public the codes. But readers can contact us to get the codes by e-mail jhong_email@163.com.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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